F26 AerE 3310: Flight Control Systems I

Practice Problem Set 2: Dynamic Response

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Problem 1

Consider the following SISO impulse response functions for linear control systems with zero initial conditions.

For each of the following, argue (1 sentence each) whether the impulse response function h corresponds to an LTI or LTV system.

(a) $h(t, \tau) = e^{-t} \cos \tau$

(b) $h(t, \tau) = t^2 + \tau^2 - 2t\tau$

(c) $h(t, \tau) = t^2 + \tau^2 + 2t\tau$

Solution.

(a) LTV system because $e^{-t} \cos \tau$ cannot be written only as a function of $t - \tau$.

(b) LTI system because $t^2 + \tau^2 - 2t\tau = (t - \tau)^2$ which is a function of the time elapsed $t - \tau$ only.

(c) LTV system because $t^2 + \tau^2 + 2t\tau$ cannot be written only as a function of $t - \tau$.

Problem 2

Consider two interconnections of a pair of causal SISO LTI impulse response functions h_1, h_2, as shown above.

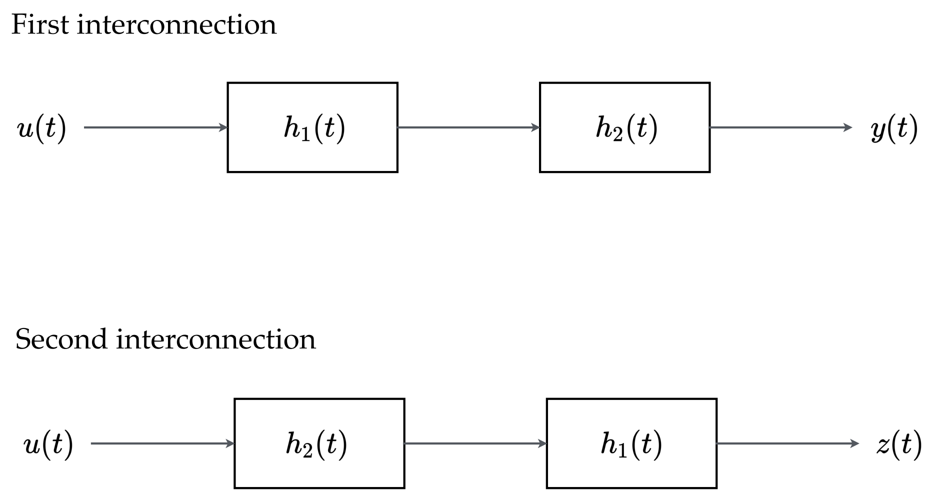

Figure 1: Two interconnections of causal SISO LTI systems.

(a) For any arbitrary input signal $u(t)$, what is the relation between the output signals $y(t)$ and $z(t)$? Explain mathematically in 1-2 sentences.

Solution for Problem 1(a)

We have $y(t) = z(t)$ because

$$y(t) = (u * h_1) * h_2 = u * (h_1 * h_2) = u * (h_2 * h_1) = (u * h_2) * h_1 = z(t),$$

where we have used the associative and commutative properties of the convolution $(*)$ operator (see Lec. 6 p. 2).

(b) For $t \geq 0$, let

$$u(t) = 1, \quad h_1(t) = e^{-t}, \quad h_2(t) = e^{-2t}.$$

Without using Laplace transform, determine $y(t), z(t)$. Your answer to part (a) should be useful. Show all your calculations.

Solution for Problem 1(b)

We have

$$(u * h_1)(t) = (h_1 * u)(t) = \int_0^t \underbrace{h_1(\tau)}_{e^{-\tau}} \underbrace{u(t-\tau)}_{1} \mathrm{d}\tau = 1 - e^{-t},$$

which using part (a), gives

$$\begin{aligned} y(t) = z(t) = ((u * h_1) * h_2)(t) = \int_0^t \left(1 - e^{-(t-\tau)}\right) e^{-2\tau} \mathrm{d}\tau &= \int_0^t \left(e^{-2\tau} - e^{-t} e^{-\tau}\right) \mathrm{d}\tau \\ &= \frac{1}{2} e^{-2t} + \frac{1}{2} - e^{-t}. \end{aligned}$$

(c) Redo part (b), i.e., compute $y(t), z(t)$, but this time using Laplace transform. Show the details of your calculations. Obviously, your final answer here should match with that in part (b).

Solution for Problem 1(c)

We have

$$\mathcal{L}[y(t)] = \mathcal{L}[z(t)] = \mathcal{L}[u(t)]\,\mathcal{L}[h_1(t)]\,\mathcal{L}[h_2(t)] = \frac{1}{s}\cdot\frac{1}{s+1}\cdot\frac{1}{s+2}.$$

By partial fraction expansion,

$$\frac{1}{s(s+1)(s+2)} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{s+2} \Rightarrow 1 = A(s+1)(s+2) + Bs(s+2) + Cs(s+1).$$

For $s = 0$, we get $A = 1/2$. For $s = -1$, we get $B = -1$. For $s = -2$, we get $C = 1/2$. Therefore,

$$\begin{aligned} y(t) = z(t) = \mathcal{L}^{-1}\left[\frac{1}{s(s+1)(s+2)}\right] &= \frac{1}{2}\mathcal{L}^{-1}\left[\frac{1}{s}\right] - \mathcal{L}^{-1}\left[\frac{1}{s+1}\right] + \frac{1}{2}\mathcal{L}^{-1}\left[\frac{1}{s+2}\right] \\ &= \frac{1}{2} - e^{-t} + \frac{1}{2}e^{-2t}, \end{aligned}$$

which is the same as what we got in part (b), as expected.